

RAJALAKSHMI ENGINEERING COLLEGE

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FINANCE AND EXPENSE ANALYSIS USING POWER BI

A Mini Project Report

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ABSTRACT

Personal finance management involves recording income, monitoring expenses, understanding savings, and checking whether spending remains within planned limits. When transaction records are reviewed manually, it can be difficult to identify spending patterns, compare categories, and understand changes over time. This mini project, titled “Finance and Expense Analysis using Power BI,” presents a visual approach to examining financial transactions and budget performance.

The Power BI report uses the FinancialData, DateTable, CategoryTable, and Budget tables, with measures including Total Income, Total Expenses, Net Savings, Savings Rate, Average Expense, Total Budget, Actual Expenses, Budget Variance, and Budget Used %. The report is organized into four pages: financial overview, expense analysis, budget analysis, and transaction details. Cards, charts, slicers, and a transaction table help users explore the available information.

The main purpose of the project is to convert transaction and budget data into an understandable report that supports monitoring and comparison. The dashboard can help identify expense categories that contribute substantially to spending, observe income and expense trends, and compare actual expenses with budget values. Findings should be interpreted according to the dataset and filters selected in Power BI.

Keywords: Finance Analytics, Expense Analysis, Microsoft Power BI, Data Visualization, Budget Monitoring, Income, Expenses, Savings.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Personal finance analysis is the process of understanding how money is received, spent, and saved. Individuals and households commonly record transactions across different categories and payment methods. Reviewing these records helps them understand spending habits and plan future expenses.

A large transaction list can be difficult to interpret by reading rows one at a time. Data visualization simplifies this process by summarizing values and displaying comparisons through cards, charts, and tables. Microsoft Power BI provides interactive report features that allow users to filter information and examine different aspects of financial data.

This project develops a Power BI report for finance and expense analysis. It uses financial transaction data to present total income, total expenses, net savings, savings rate, category-wise expenses, monthly trends, and budget comparisons. The report is designed to make the information easier to understand and explore.

1.2 Problem Statement

Financial transaction records may contain many dates, descriptions, categories, amounts, and payment methods. Manually examining these records makes it difficult to compare expenses, understand monthly changes, identify major spending categories, and determine whether actual spending exceeds the planned budget. A visual reporting solution is therefore needed to summarize financial data and present the results in a clear and interactive form.

1.3 Objectives

- To summarize income, expenses, net savings, and savings rate.
- To analyze expenses by category and payment method.
- To examine income and expense changes across time.

- To compare budget values with actual expenses.
- To identify budget variance and budget utilization.
- To present financial information using interactive Power BI visuals.
- To provide a report that supports clearer review of transaction data.

1.4 Scope of the Project

The project focuses on the financial records and budget information available in the Power BI file. It includes high-level financial indicators, category-based expense comparisons, time-based analysis, budget monitoring, and transaction-level details. The report does not automatically guarantee financial advice or predict future outcomes; its usefulness depends on the completeness, correctness, and date range of the underlying data.

1.5 Organization of the Report

Chapter 1 introduces the project. Chapter 2 explains the tools and technologies. Chapter 3 describes the data model and key fields. Chapter 4 explains the methodology. Chapter 5 describes the dashboard pages and visualizations. Chapter 6 discusses how to interpret the results. Chapter 7 presents advantages, limitations, and future enhancements. Chapter 8 provides dashboard output documentation, and Chapter 9 concludes the project.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Microsoft Power BI

Microsoft Power BI is a business intelligence and data visualization platform used to transform structured data into interactive reports. It provides cards, charts, tables, slicers, formatting options, and data modeling features. In this project, Power BI is used to display financial measures, compare expense categories, track budget performance, and explore transaction details.

2.2 Power Query and Data Preparation

Power Query is the data preparation environment available in Power BI. It can be used to inspect columns, set data types, rename fields, handle missing values, remove duplicate records where appropriate, and prepare data before it is loaded into the report model. Any cleaning operation should preserve valid financial transactions and should be documented.

2.3 DAX Measures

Data Analysis Expressions (DAX) is the formula language used in Power BI to create calculated measures and columns. The report contains measures named Total Income, Total Expenses, Net Savings, Savings Rate, Average Expense, Total Budget, Actual Expenses, Budget Variance, and Budget Used %. These measures summarize the available data and are used by the report visuals.

2.4 Data Model Tables

The report definitions refer to the tables FinancialData, DateTable, CategoryTable, and Budget. FinancialData contains transaction-oriented fields such as Date, Amount, Description, Category, PaymentMethod, and Type. DateTable supports time-based reporting, CategoryTable provides category labels, and Budget contains budget-related measures and values.

2.5 Summary of Technologies

Technology	Purpose
Microsoft Power BI	Report design and interactive visualization
Power Query	Data inspection and transformation
DAX	Financial calculations and measures
FinancialData table	Transaction and payment details
DateTable	Date and month-based analysis
CategoryTable	Expense category analysis
Budget table	Budget and actual expense comparison

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The Power BI report is built around financial transaction and budget information. The report model references the FinancialData, DateTable, CategoryTable, and Budget tables. Transaction fields allow analysis by date, category, payment method, type, description, and amount. Budget-related fields allow planned values to be compared with actual expenses.

3.2 Main Transaction Attributes

Field	**Description / Use**
Date	Date associated with a financial transaction
Amount	Monetary value recorded for a transaction
Description	Text describing the transaction
Category	Category associated with the transaction
PaymentMethod	Method used to make or receive a transaction
Type	Transaction type, such as income or expense, according to the dataset coding
MonthYear	Month and year label used for time-based visuals

3.3 Financial Measures

Measure	**Purpose**
Total Income	Summarizes income values
Total Expenses	Summarizes expense values
Net Savings	Represents the difference between income and expenses, according to the implemented measure
Savings Rate	Expresses savings relative to a defined base, according to the implemented formula

Average Expense	Shows the average expense according to the measure definition
Total Budget	Summarizes planned budget values
Actual Expenses	Summarizes actual expenses used in budget comparison
Budget Variance	Shows the difference between budget and actual expense values
Budget Used %	Shows budget utilization as a percentage

3.4 Data Quality Considerations

Before drawing conclusions, transaction amounts and dates should use appropriate data types. Category and payment-method labels should be consistent. Missing or duplicated records should be examined before removal because repeated-looking transactions may be legitimate. The sign convention for income and expense amounts, as well as the formulas used by the measures, should be verified in Power BI.

CHAPTER 4: METHODOLOGY

4.1 Project Workflow

The project follows a sequence from data preparation to interactive reporting: financial data and budget information → data inspection → cleaning and transformation → data modeling and measure creation → visual design → validation → interpretation of results.

4.2 Data Inspection and Preparation

The data preparation stage includes checking table and column names, verifying data types, reviewing date values, and inspecting category and payment-method labels. Where required, Power Query can be used to transform columns and prepare a consistent dataset. The report should retain a clear distinction between income transactions and expense transactions.

4.3 Data Modeling

A date table supports grouping by month and time period. Category labels support grouping transactions by category. The Budget table supports comparison of budget values with actual expenses. Relationships between tables should be checked so that slicers and charts filter the intended data consistently.

4.4 Measure-Based Analysis

The report uses measures for income, expenses, savings, savings rate, average expense, budget, actual expenses, variance, and utilization. The exact calculation of each measure should be confirmed in the model. For example, net savings is commonly calculated as total income minus total expenses, but the implemented DAX formula should be used when reporting final values.

4.5 Dashboard Design and Validation

The report organizes the measures and fields into four pages. Cards highlight summary indicators, charts compare categories and time periods, slicers support

filtering, and the transaction table provides detail. Before submission, each visual should be tested with slicers, totals should be checked against the underlying data, and titles and number formats should be reviewed.

CHAPTER 5: POWER BI

DASHBOARD AND VISUALIZATIONS

5.1 Dashboard Overview

The uploaded Power BI file contains four report pages. The visuals use the financial measures and fields from the report model. The following descriptions document the purpose of each page based on the visual field references in the PBIX file.

5.2 Page 1: Financial Overview

The first page includes summary cards for Total Income, Total Expenses, Net Savings, and Savings Rate. It also includes a time-based comparison using MonthYear and income/expense measures, as well as category and date filtering controls.

The summary cards provide a quick view of overall financial position. The monthly comparison helps users examine whether income and expenses change over time. Category-based visuals help identify where expenses are concentrated. When reviewing this page, users should select an appropriate date range and confirm that the values reflect the intended filters.

5.3 Page 2: Expense Analysis

The second page includes measures such as Total Income, Total Expenses, and Average Expense, together with category-related comparisons, time-based visuals, and filters for category, payment method, and transaction type. These visuals are intended to help users compare spending across categories and understand the distribution of transactions.

A category chart helps identify categories with higher recorded expenses. A payment-method visual provides a way to compare transaction totals by payment method. Monthly expense views can reveal periods with higher or lower spending.

These patterns describe the available data and do not by themselves explain why a particular expense occurred.

5.4 Page 3: Budget Analysis

The third page contains Total Budget, Actual Expenses, Budget Variance, and Budget Used % measures. It includes budget comparison visuals and a table with category and transaction-related details. This page is designed to compare planned budget values with actual expenses and to identify categories that may need attention.

Budget variance must be interpreted using the sign convention in the DAX measure. If variance is defined as budget minus actual expenses, a negative result may indicate overspending. If it is defined as actual expenses minus budget, a positive result may indicate overspending. Budget utilization should also be checked to confirm whether it is displayed as a decimal or percentage.

5.5 Page 4: Transaction Details

The fourth page includes transaction-level fields such as Date, Amount, Description, Category, PaymentMethod, and Type, along with filters. The table supports examination of individual records behind the aggregated totals. Users can use this page to inspect transactions after filtering by date, category, payment method, or transaction type.

5.6 Interactive Features

Slicers allow the report to be filtered by available dimensions. When configured correctly, a selection on one page or visual can affect other visuals according to the report interaction settings. Tooltips, drill-through pages, and navigation buttons may also be used to make the report easier to explore and present.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results Presented by the Dashboard

The dashboard provides a consolidated view of income, expenses, net savings, savings rate, category-wise spending, monthly changes, and budget performance. The exact numeric results depend on the dataset loaded in the PBIX file and the filters selected at the time of viewing. Numeric totals have therefore not been invented in this report.

The overview page supports a quick review of the relationship between income and expenses. The expense analysis page supports category and payment-method comparisons. The budget page highlights the relationship between planned budgets and actual expenses. The transaction page allows users to inspect individual records to understand the details behind aggregated values.

6.2 Discussion

Using multiple financial measures is more informative than relying on a single total. High expenses do not necessarily indicate poor financial management unless they are considered alongside income, the selected period, category needs, and budget limits. Similarly, a savings rate is meaningful only when its formula and denominator are understood.

The report's interactive design allows users to explore a question from different perspectives. For example, a user can identify a category with high expenses and then filter by month or payment method to inspect the pattern. Budget variance can direct attention to categories that may require review. These are descriptive insights and should be interpreted in the context of the available records.

6.3 Validation of Findings

Before final submission, record the exact totals displayed by Power BI, check the date range, and verify key measures against transaction records. Confirm that the

budget and actual values are compared at the same category and time granularity. Any screenshots and numerical findings added to the final report should match the final saved PBIX file.

CHAPTER 7: ADVANTAGES, LIMITATIONS AND FUTURE ENHANCEMENTS

7.1 Advantages

- Summarizes important financial indicators in one report.
- Makes it easier to compare income and expenses over time.
- Supports category-wise and payment-method analysis.
- Provides budget and actual expense comparisons.
- Allows users to filter the report and inspect transaction details.
- Presents financial data visually instead of relying only on raw tables.

7.2 Limitations

- The accuracy of results depends on the completeness and quality of the source data.
- The report describes recorded transactions and cannot independently verify whether they are correct.
- Budget conclusions depend on the budget values and variance formula used in the model.
- Historical spending patterns do not guarantee future income or expenses.
- The report is an analysis tool and does not replace professional financial advice.

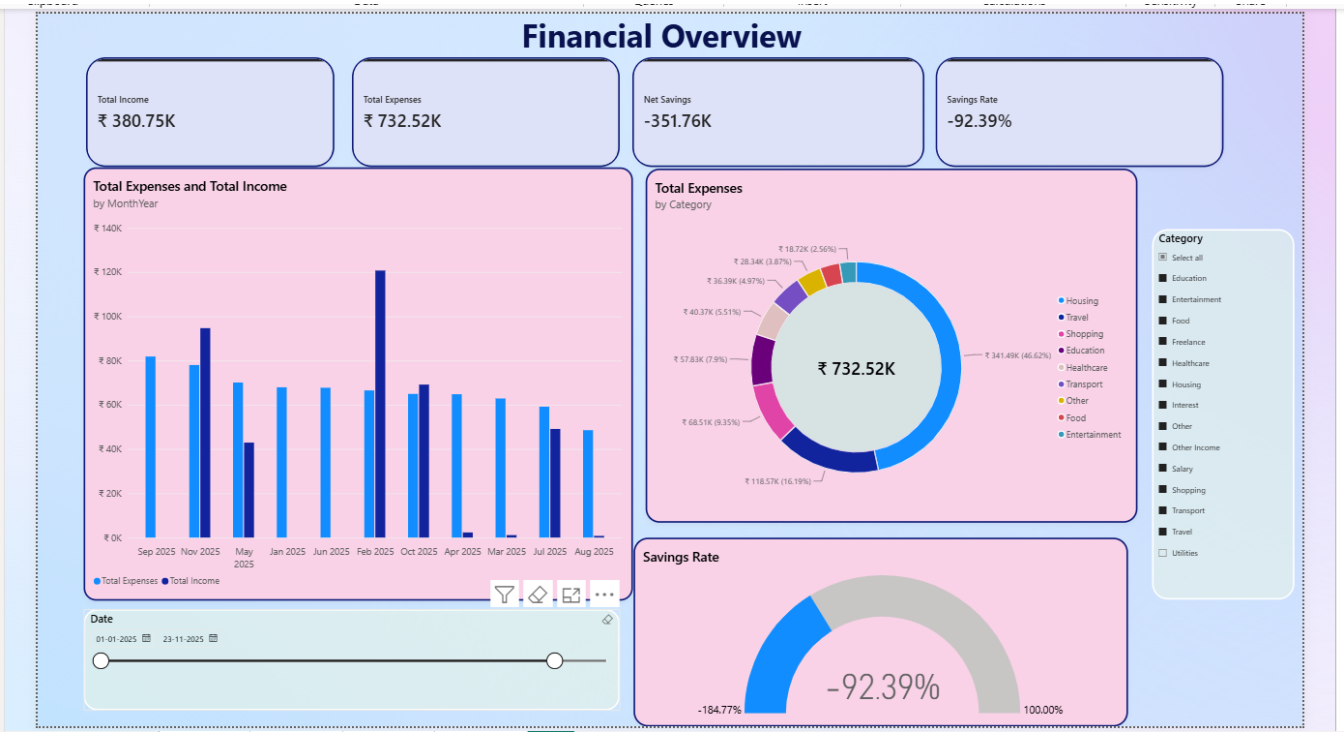
7.3 Future Enhancements

- Add a savings goal tracker and monthly savings target.
- Add alerts for categories that exceed a defined budget threshold.
- Add year-over-year and month-over-month comparisons where sufficient dates are available.
- Connect to a refreshed data source if the project requires regular updates.
- Add documentation of the DAX formulas and data cleaning rules.

CHAPTER 8: POWER BI DASHBOARD OUTPUT

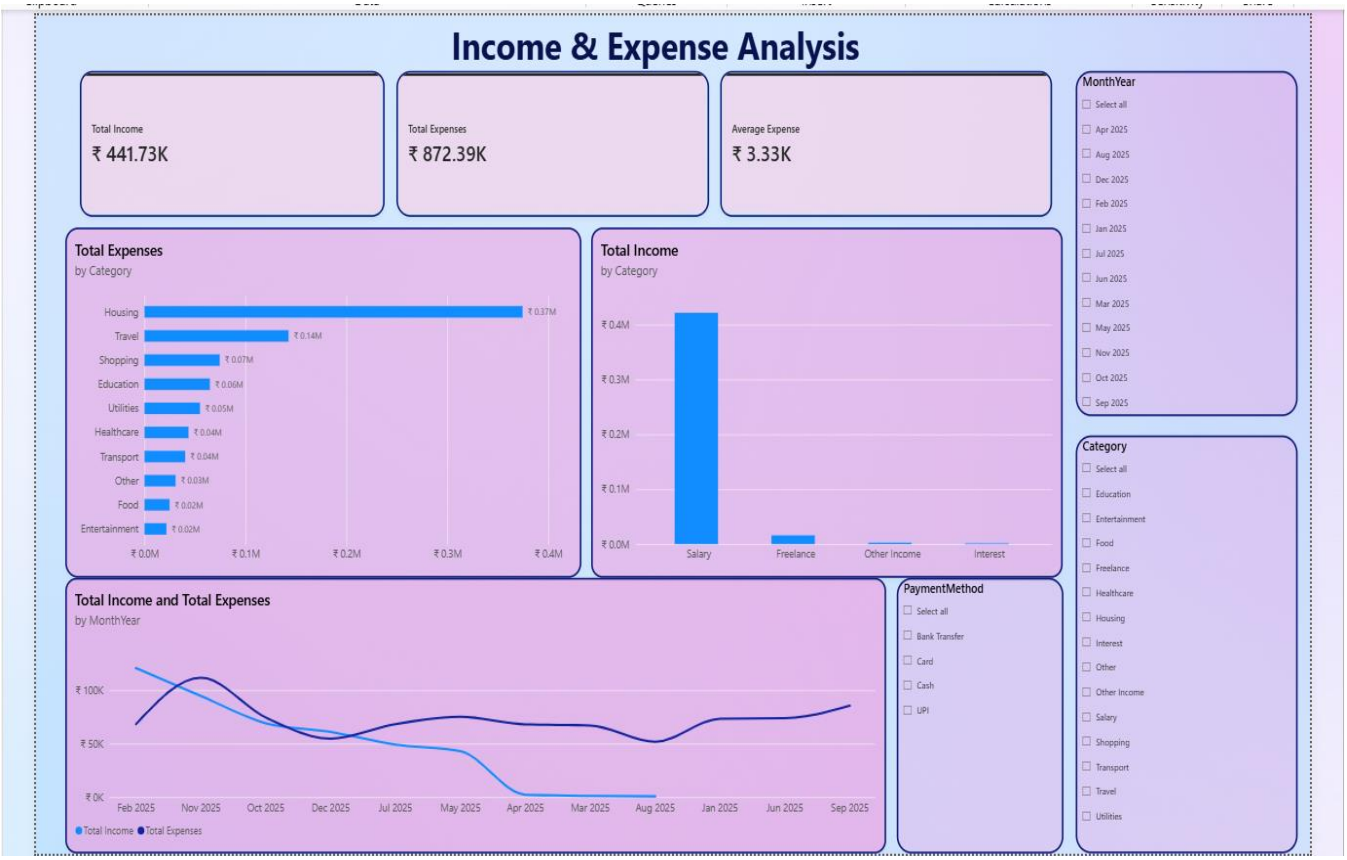
This chapter is intended to document the final Power BI report output. Insert clear screenshots exported from the final saved PBIX file in the spaces below. Ensure that screenshots show the full report page, readable chart titles, and the filters used when interpreting any displayed values.

8.1 Page 1: Financial Overview



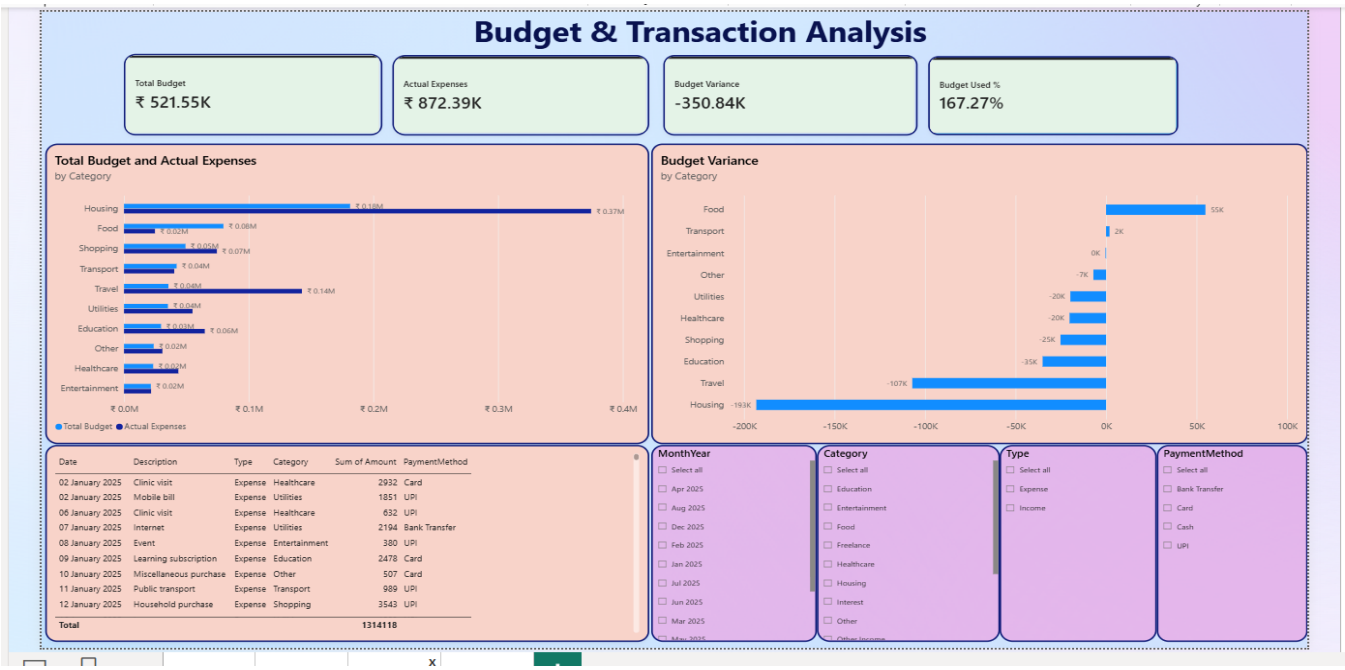
It shows the income, expense, net savings, and savings-rate indicators along with the time-based and category-based visuals.

8.2 Page 2: Expense Analysis



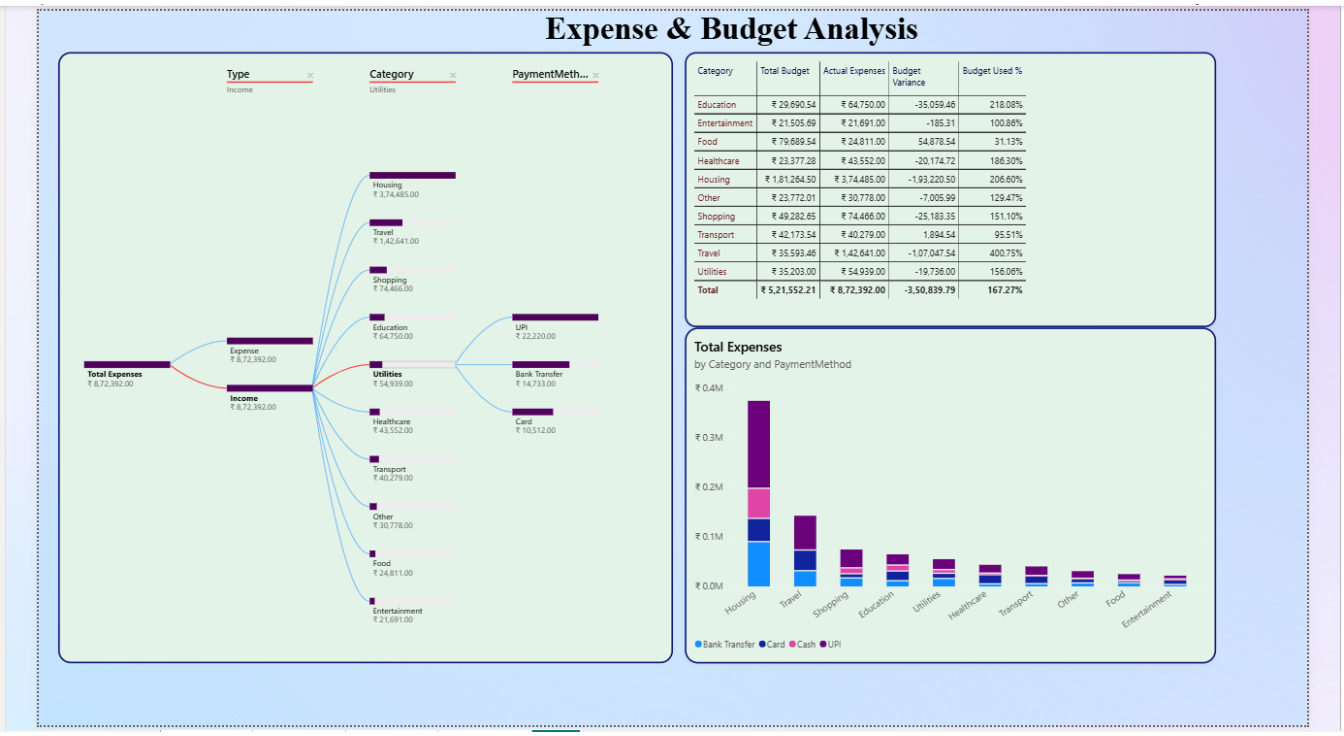
It shows the expense analysis visuals, category comparisons, and any relevant payment-method or date filters.

8.3 Page 3: Budget Analysis



It shows budget, actual expenses, variance, utilization, and category-level comparison details.

8.4 Page 4: Transaction Details



It shows the transaction table and relevant filters. Hide or mask any personal or sensitive transaction details before sharing the report.

CHAPTER 9: CONCLUSION

The Finance and Expense Analysis mini project demonstrates how Microsoft Power BI can be used to organize financial transaction and budget information into an interactive report. The report model includes measures for income, expenses, net savings, savings rate, average expense, budget, actual expenses, variance, and budget utilization.

The four report pages provide complementary views of the data: a financial overview, expense analysis, budget analysis, and transaction details. Together, these views support review of overall financial position, comparison of expense categories, observation of changes over time, and monitoring of budget performance.

The project highlights the value of data visualization in making financial information easier to explore. The reliability of any conclusion depends on accurate source data, correct measure definitions, valid relationships, and appropriate filters. Future enhancements such as budget alerts, savings targets, and time-based comparisons could make the report more useful for ongoing financial monitoring.

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4. Uploaded Power BI project file: mini project.pbix.